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Heritage Guard

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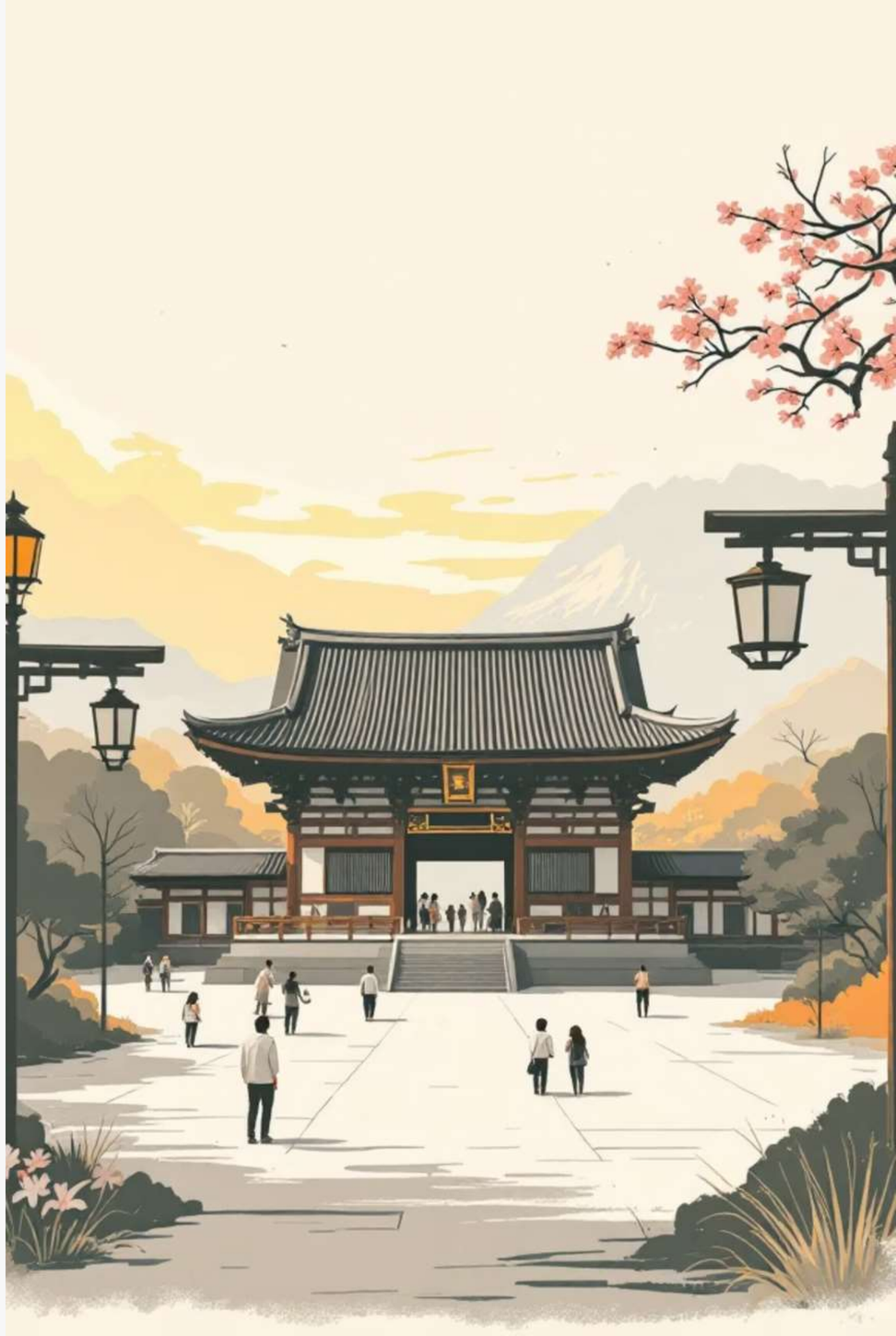
Preserving Our Past: An AI-Driven Approach to Cultural Heritage Conservation

Cultural heritage sites are irreplaceable symbols of human history and identity — yet they are under growing threat. This presentation outlines a unified, intelligent system to protect them for future generations.



The Challenge We Face

Over-Tourism

High visitor traffic accelerates wear on fragile structures and artifacts, creating tension between access and preservation.

Pollution

Airborne contaminants and environmental degradation cause irreversible chemical and physical damage to heritage materials.

Climate Change

Rising temperatures, flooding, and extreme weather events increasingly threaten the structural integrity of historic sites.

Why Current Approaches Fall Short

The current landscape breaks down in three distinct ways:

Fragmented Systems

Monitoring, restoration, and tourism tools work in silos, blocking coordinated action.

Reactive Management

Most systems act only after damage is done, missing the prevention window.

Lost Authenticity

Without continuous monitoring, slow deterioration goes unnoticed until it’s irreversible.

The gap: A unified, intelligent platform is needed to connect these failures.

What AI Can Do — But Doesn't Yet

Current Capabilities

- **ML models** predict tourist flow and crowd patterns
- **Deep learning** restores damaged images and artifacts
- **NLP** analyzes historical texts for cultural validation
- **Computer vision** detects structural issues in real time

The Critical Gap

Each capability exists in isolation. No single platform integrates crowd prediction, restoration, authenticity analysis, and real-time monitoring into a cohesive decision-support system for heritage managers.

Problem Statement

No unified AI-based platform currently exists that can address the full spectrum of heritage preservation challenges simultaneously.

Predict Damage

Anticipate structural and environmental risks before they occur

Manage Tourist Flow

Balance public access with long-term site preservation

Assist Restoration

Digitally reconstruct and guide physical repair efforts

Maintain Authenticity

Ensure cultural integrity is preserved through every intervention

Research Objectives

1

Image Restoration Module

Use GAN models to digitally reconstruct damaged heritage elements with high fidelity.

2

Crowd Prediction System

Analyse and forecast visitor patterns using machine learning to enable proactive management.

3

Authenticity Analysis

Evaluate cultural data and historical records using NLP techniques to generate authenticity scores.

4

Interactive Dashboard

Deliver real-time visualization and actionable insights through a user-friendly frontend interface.

Proposed Methodology

PHASE 1: DATA COLLECTION & PROCESSING

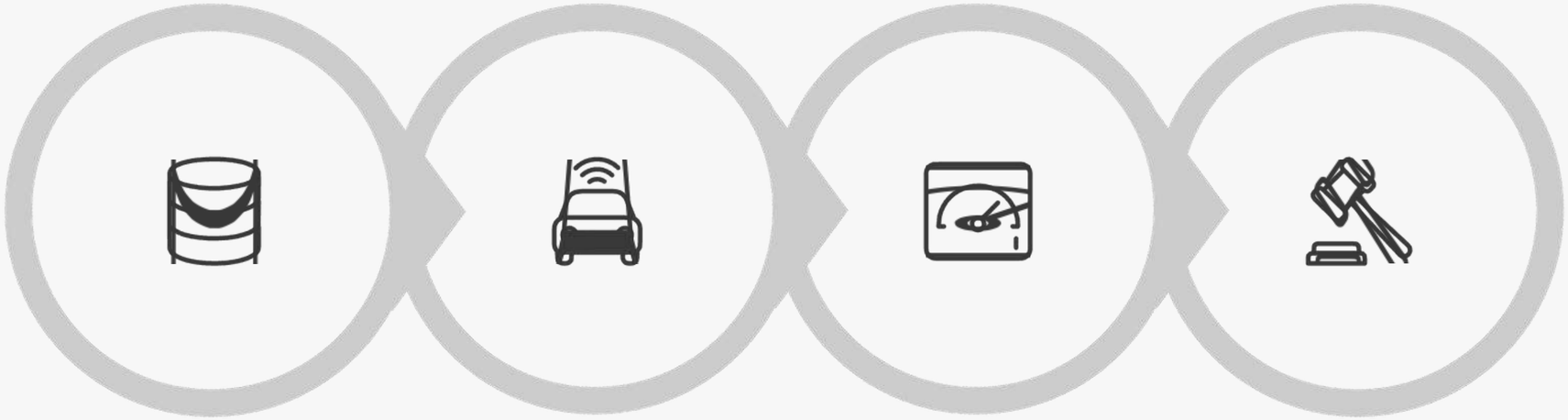
Data Sources

- Heritage images & structural reports
- Historical documents & texts
- Tourism & visitor data
- Environmental sensor data

Processing Pipeline

- Image enhancement — noise removal and color correction
- Text cleaning and standardization for NLP input
- Feature extraction to prepare data for ML models

AI System Framework



Collect Process Visualize Act

The platform unifies three core AI components — an NLP module for authenticity scoring, a GAN-based restoration system, and a real-time visualization layer — into a single decision-support pipeline for heritage managers.

Expected Outcomes & Project Progress

Expected Outcomes

Accurate Predictions

Crowd and environmental impact forecasting

Digital Restoration

AI-generated previews of repaired structures

Authenticity Scores

Historical analysis-based cultural integrity metrics

Centralized Dashboard

Unified monitoring and decision-making interface

Progress to Date

- Initial research draft completed
- Basic dashboard prototype developed
- Core system architecture designed

Next Steps

- Deploy on selected heritage pilot sites
- Enhance real-time monitoring capabilities
- Refine UI/UX for field usability

Conclusion: A Modern Solution for an Ancient Responsibility

This project delivers a unified, AI-driven platform that brings together crowd prediction, digital restoration, authenticity analysis, and real-time monitoring — transforming how we protect the irreplaceable.

Efficient	Sustainable
Proactive, data-informed decisions replace reactive responses	Long-term preservation balanced with responsible public access

Unified

One intelligent platform replaces fragmented, siloed tools